

RESULTS

Results

01 · WILCOXON SIGNED-RANK

nonparametric paired comparison

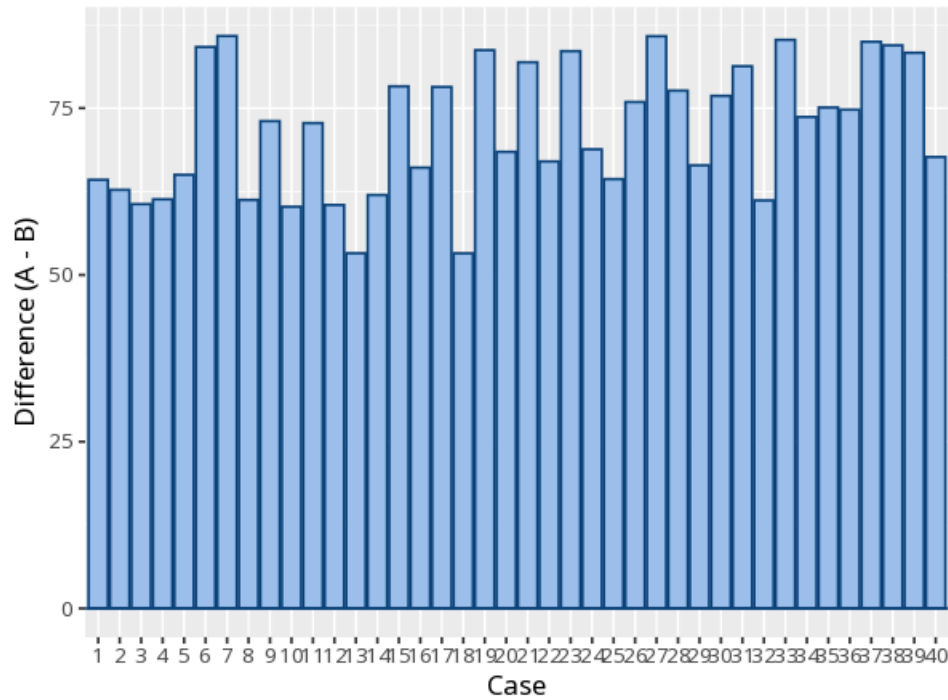
Table 1. Rank summary

Sign	N	Mean rank	Sum of ranks
Positive	40	20.50	820.00
Negative	0	—	—
Ties	0	—	—

Table 2. Signed-rank test

V / W	Z	p	r [95% CI]	Median diff [95% CI]
820.00	5.51	<.001	1.00 [1.00, 1.00]	72.24 [68.63, 75.10]

Figure. Change per case



Understanding the numbers

Hodges–Lehmann median diff. The estimated median of the paired differences (a location-shift estimate), with its confidence interval. Here, the Hodges–Lehmann median difference is 72.24 [68.63, 75.10].

Mean rank. The average rank of the absolute differences within the positive- and negative-change groups. Here, the mean rank is 20.50 for positive changes and — for negative changes.

N. How many cases showed a positive change, a negative change, or no change (a tie). Here, N = 40 positive, 0 negative, and 0 tied.

p. The probability of a shift this large (or larger) if there truly were no systematic difference between conditions. Here, $p < .001$ - below the conventional .05 threshold, a significant shift.

r. The rank-biserial effect size for the shift between the two conditions. Here, $r = 1.00$ [1.00, 1.00].

Sign. Whether each case's change was positive, negative, or tied (no change). Here, this run's cases split into 40 positive, 0 negative, and 0 tied changes.

Sum of ranks. The total of the ranks assigned to the positive and negative differences – the test statistic V is derived from this. Here, the sum of ranks is 820.00 for positive changes and — for negative changes.

V. The Wilcoxon signed-rank test statistic, the smaller of the summed positive/negative ranks. Here, $V = 820.00$.

Z. The standardized test statistic used to compute p. Here, $Z = 5.51$.

How to read this test

The nonparametric counterpart to the paired t-test. It ranks the size of each paired change; a p below alpha means a systematic shift between the two conditions, with r as the effect size. The shift maps to the median difference only when the within-pair differences are roughly symmetric; the sign of the median difference (or Hodges–Lehmann estimate) gives the direction. Your significance threshold (α) is 0.05. Method: Wilcoxon signed rank exact test.

APA template: A Wilcoxon signed-rank test gave $V=820.00$, $Z=5.51$, $p < .001$, $r=1.00$ [1.00, 1.00].

Statistical basis: Wilcoxon signed-rank test (Wilcoxon, F. (1945). "Individual comparisons by ranking methods." Biometrics Bulletin, 1(6), 80-83.); Matched-pairs rank-biserial effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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